

SOC Home Lab: Sysmon & Splunk

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Introduction

Modern Security Operations Centers (SOCs) rely on endpoint monitoring and centralized log management to detect, investigate, and respond to security incidents. Security analysts use tools that collect and analyze large volumes of system activity to identify suspicious behaviour and potential threats.

In this project, I built a small home lab to learn more about Security Operations Centre (SOC) activities and log analysis. The lab consisted of a Windows 10 Pro virtual machine running in VirtualBox and Splunk Enterprise installed on my laptop.

I installed Sysmon on the Windows virtual machine and used the SwiftOnSecurity Sysmon configuration to generate detailed security logs. I also installed the Splunk Universal Forwarder on the virtual machine so that the logs could be sent to Splunk Enterprise for analysis.

The purpose of this project was to gain hands-on experience with tools commonly used by SOC analysts and to better understand how security monitoring works in real-world environments.

Aim of the Project

The aim of this project was to:

- Build a simple SOC home lab.
 - Learn how logs are generated and collected.
 - Understand how Sysmon monitors activity on a Windows system.
 - Learn how Splunk Enterprise can be used to analyse logs.
 - Gain hands-on experience with security monitoring tools.
 - Develop practical skills that are relevant to a SOC analyst role.
-

Log Analysis

Log analysis is the process of reviewing and investigating logs generated by computers, applications, and network devices.

Whenever an action takes place on a system, such as opening a program or connecting to the internet, a log can be created. These logs provide useful information about what is happening on a device.

Security analysts use logs to:

- Monitor activity
- Detect suspicious behaviour
- Investigate incidents
- Troubleshoot problems

Without logs, it would be very difficult to understand what is happening on a system.

Why is Log Analysis Important?

Log analysis is important because it helps organisations identify potential security threats and investigate incidents.

Some benefits include:

- Detecting suspicious activity
- Investigating security incidents
- Monitoring systems and users
- Identifying errors and issues
- Meeting compliance requirements

Logs provide visibility into a system and help analysts understand what actions have taken place.

Sysmon

Sysmon (System Monitor) is a Microsoft Sysinternals tool that records detailed information about activity on a Windows system.

Compared to normal Windows logs, Sysmon provides much more detailed information that can be useful during investigations.

Some events that Sysmon can monitor include:

- Process creation
- Process termination
- Network connections
- Registry changes
- Driver loading
- File creation activity

These events are stored in the Windows Event Log, and can be viewed in the application Event Viewer, it can also be collected by other tools such as Splunk.

SwiftOnSecurity Sysmon Configuration

The SwiftOnSecurity Sysmon configuration is a popular Sysmon configuration file used by many security professionals.

The default Sysmon installation can generate a large number of events, which can make analysis difficult. The SwiftOnSecurity configuration helps reduce unnecessary logs while still collecting important security information.

I used this configuration because it is widely recommended and provides useful visibility into system activity.

Splunk Enterprise

Splunk Enterprise is a SIEM (Security Information and Event Management) platform used to collect, store, and analyse logs.

Splunk allows users to search through large amounts of log data and investigate events quickly.

Some common uses of Splunk include:

- Log collection
 - Security monitoring
 - Threat detection
 - Incident investigation
 - Dashboard creation
-

Splunk Universal Forwarder

The Splunk Universal Forwarder is a lightweight application that sends logs from one machine to a Splunk server.

In this project, the Universal Forwarder was installed on the Windows 10 virtual machine.

Its job was to:

- Collect Sysmon logs
- Send them to Splunk Enterprise
- Allow the logs to be analysed centrally

Without the Universal Forwarder, Splunk Enterprise would not receive the Sysmon logs from the virtual machine.

Tools Used

Tool	Purpose
Windows 10 Pro	Virtual machine being monitored
VirtualBox	Virtualisation software
Sysmon	Generates security logs
SwiftOnSecurity Configuration	Improves Sysmon logging
Splunk Universal Forwarder	Sends logs to Splunk
Splunk Enterprise	Log analysis and monitoring
Event Viewer	Viewing Windows logs

Lab Setup

The lab consisted of:

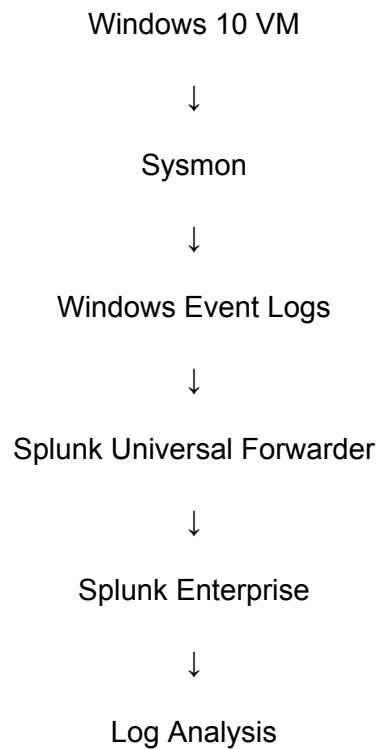
Host Machine

- Splunk Enterprise installed
- Used to search and analyse logs

Windows Virtual Machine

- Windows 10 Pro
- Sysmon installed
- SwiftOnSecurity configuration applied
- Splunk Universal Forwarder installed

Log Flow



Why I Built This Home Lab

I built this lab because I wanted hands-on experience with tools used by SOC analysts.

While studying cybersecurity concepts is important, I wanted practical experience working with security logs and monitoring tools.

This project allowed me to:

- Work with real security logs
 - Learn how SIEM systems operate
 - Understand endpoint monitoring
 - Improve my investigation skills
 - Gain experience that can help prepare me for a SOC analyst role
-

Steps I took

To complete this project:

1. Created a Windows 10 virtual machine in VirtualBox

Figure 1. Downloading Windows 10

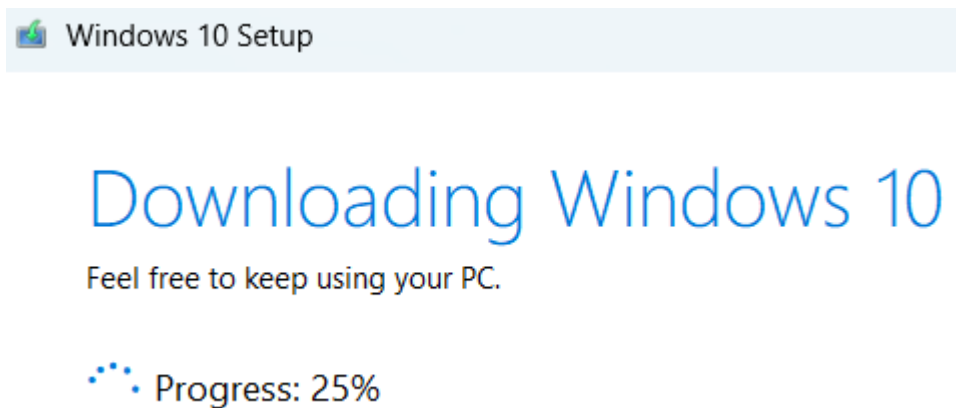


Figure 2. Installation of Windows

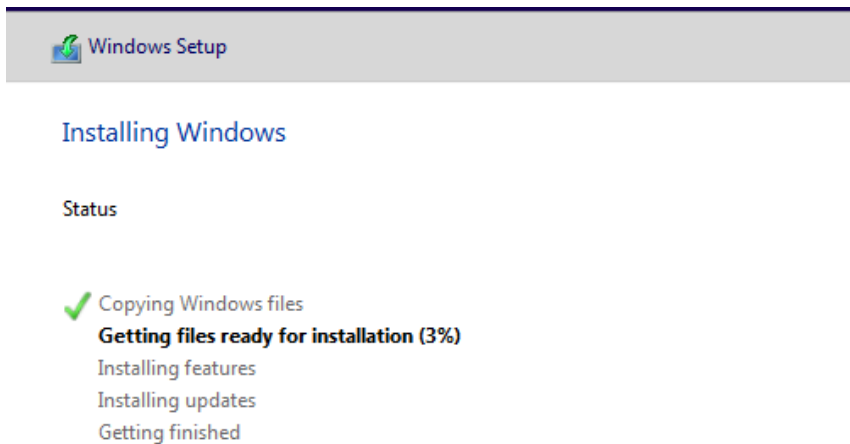
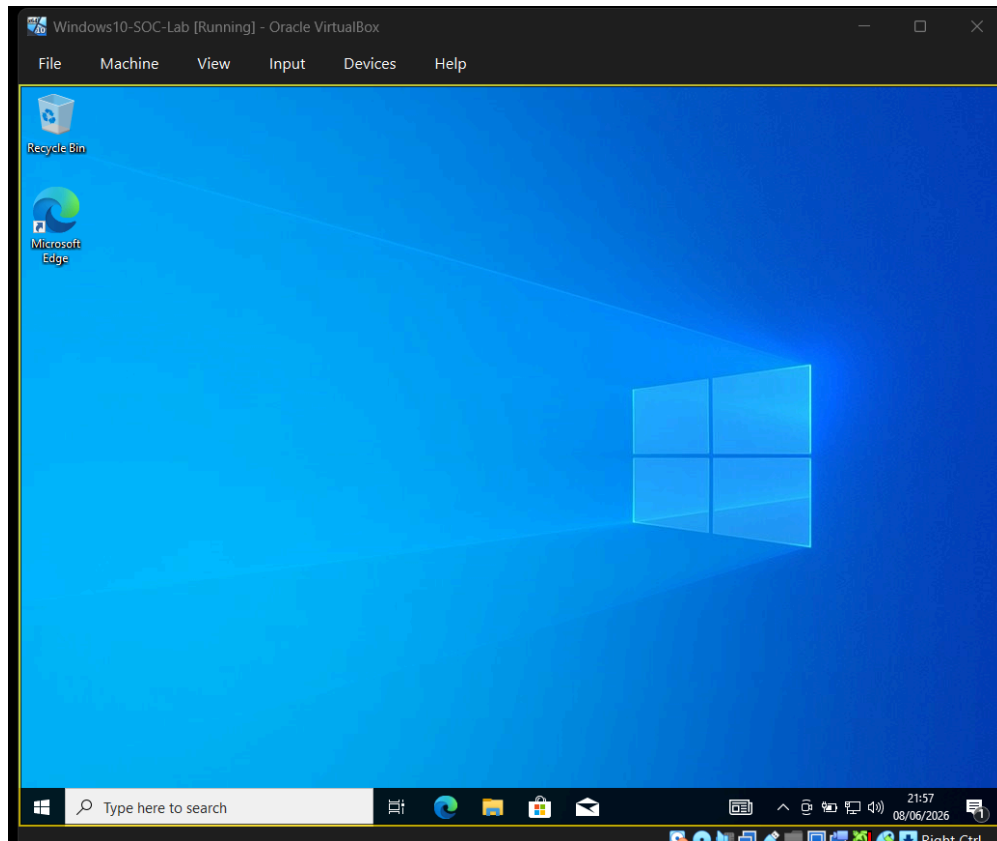
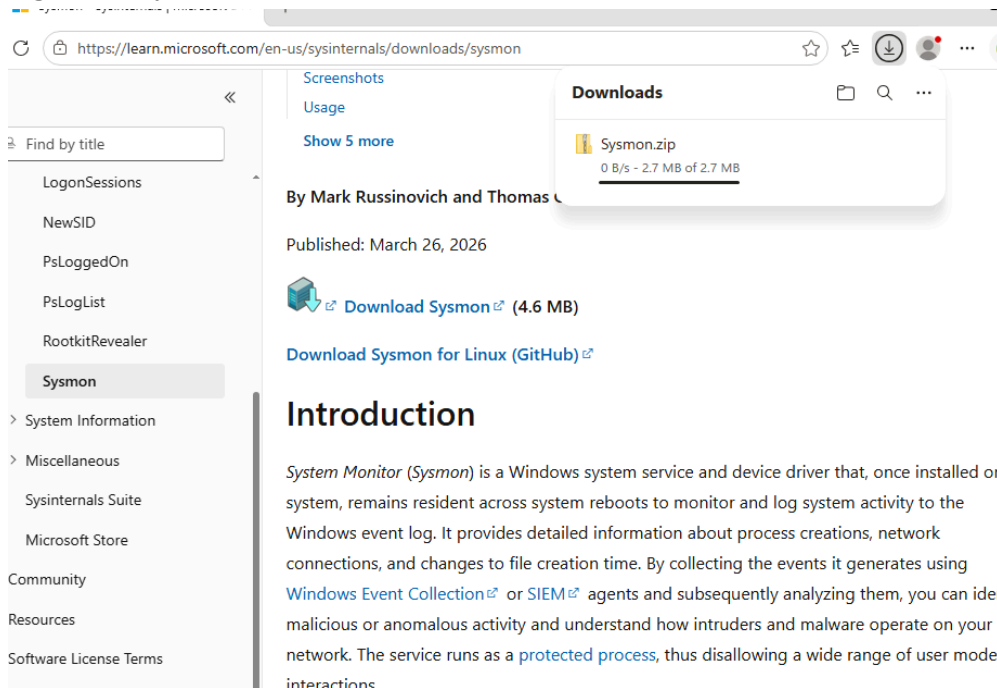


Figure 3. Windows 10 Pro successfully installed



2. Installed Sysmon on the virtual machine

Figure 4. Sysmon download



3. Applied the SwiftOnSecurity Sysmon configuration

Figure 5. SwiftOnSecurity within the same folder where Sysmon was downloaded

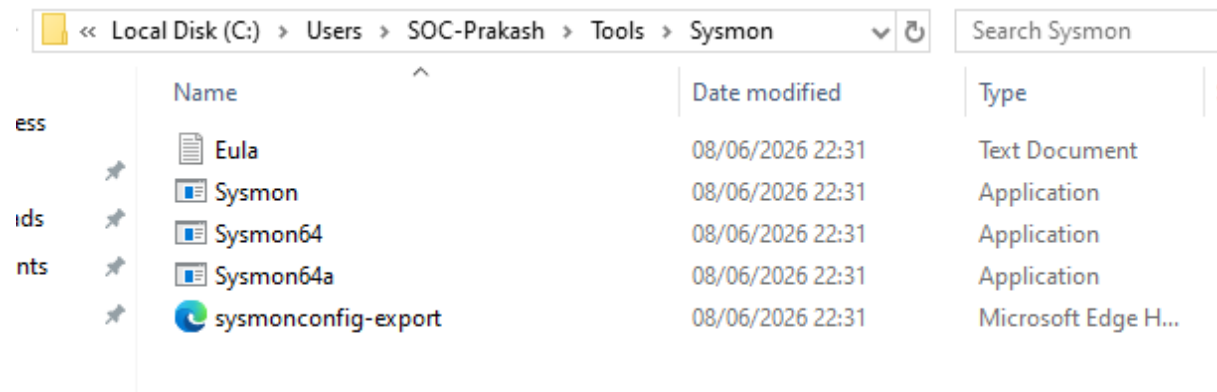


Figure 6. Installing Sysmon with PowerShell

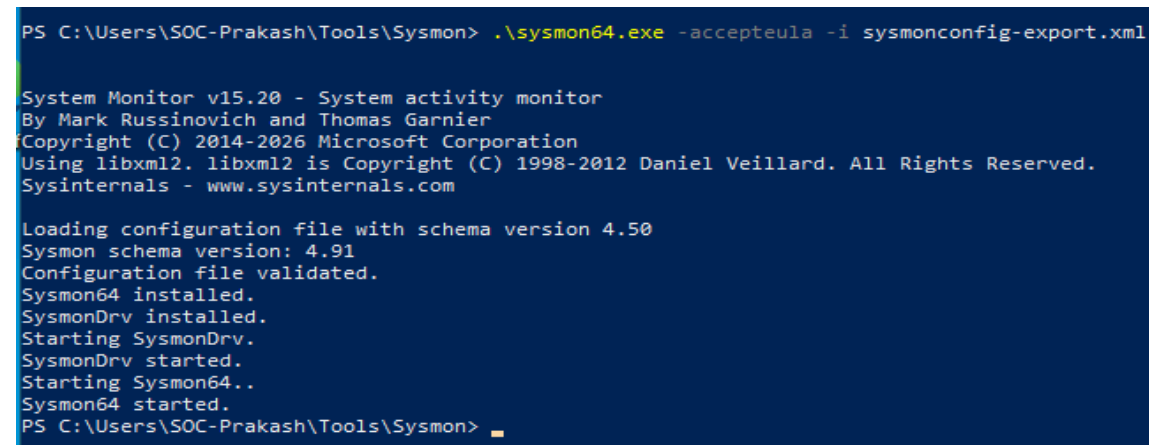


Figure 7. An overview of Event Manager

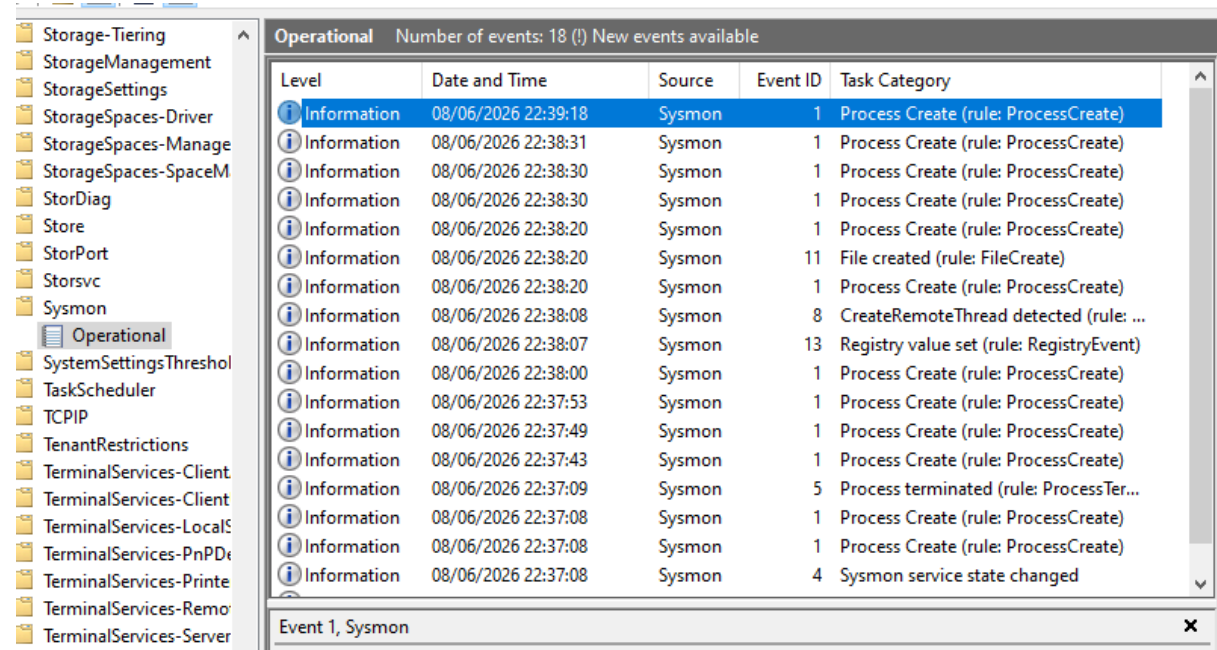
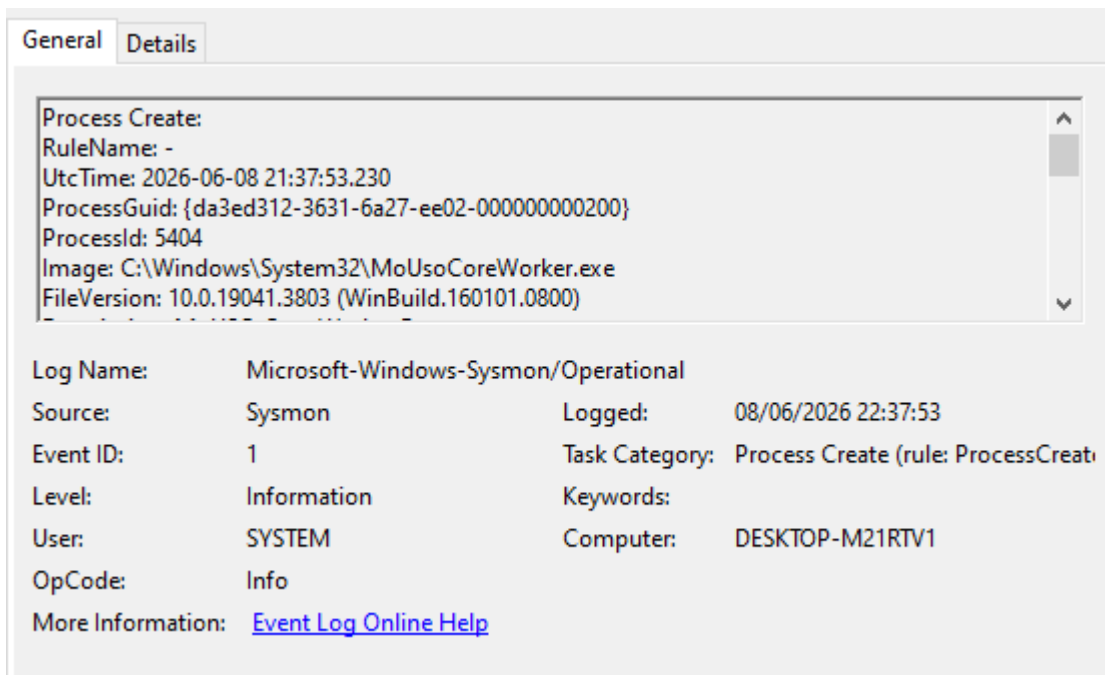


Figure 8. Process Creation Event



I generated PowerShell activity and validated that Sysmon Event ID 1 captured process creation events.

4. Installed Splunk Enterprise on my laptop

Figure 9. Downloading Splunk

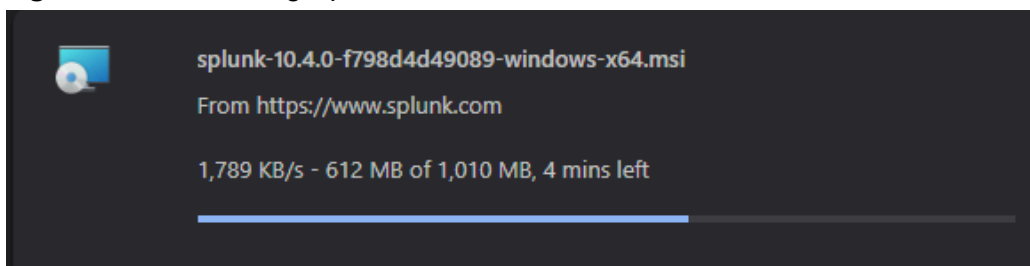


Figure 10. Splunk successfully installed

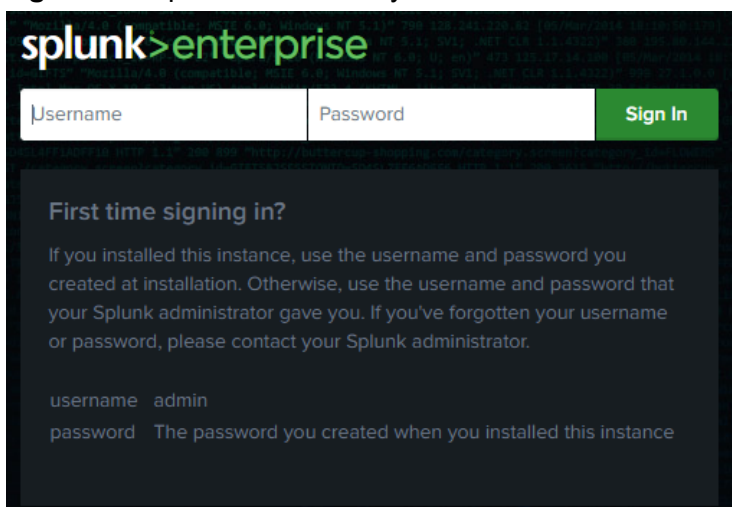


Figure 11. Configured the receiving end for Splunk Enterprise

Configure receiving

Set up this Splunk instance to receive data from forwarder(s).

Listen on this port *

9997

For example, 9997 will receive data on TCP port 9997.

5. Adding indexes

Figure 12. Index creation for searching

Add new index

* Name
sysmon

* Index data type
☒ Events ☐ Metrics
The type of data to store (event-based or metrics)

Home path

Hot/warm db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/db).

Cold path

Figure 13. Three additional indexes were created: sysmon, wineventlog & endpoint

endpoint	● Ad
history	● Ad
main	● Ad
splunklogger	● In
summary	● Ad
sysmon	● Ad
wineventlog	● Ad

6. Installed the Splunk Universal Forwarder on the virtual machine

Figure 14. Splunk Forwarder downloading

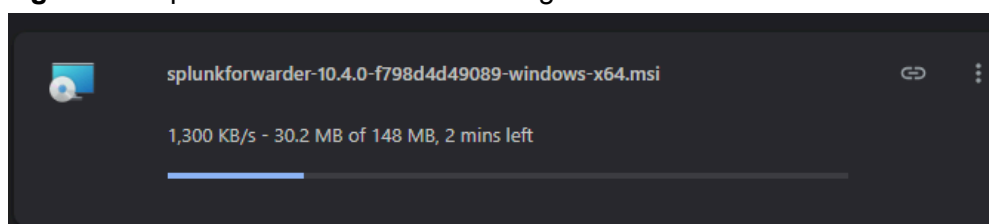
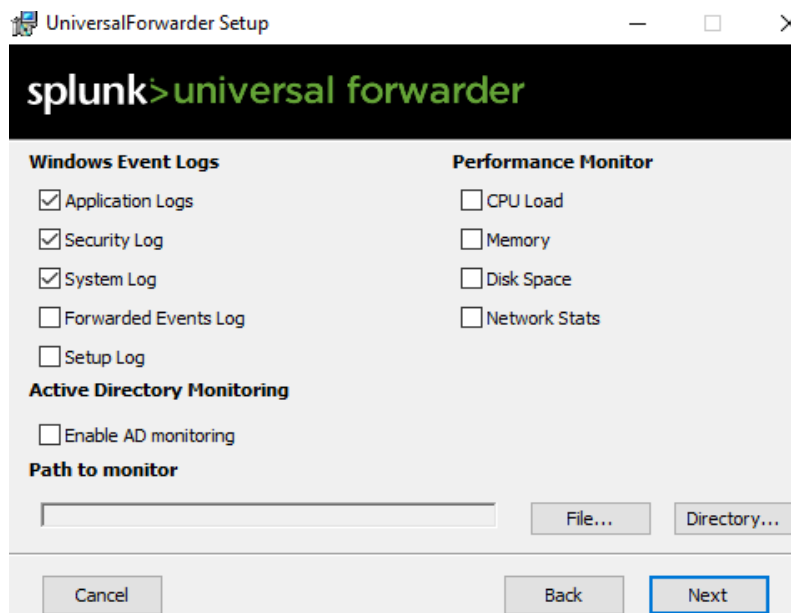


Figure 15. Setting up Splunk Forwarder



7. Configured the Universal Forwarder to send Sysmon logs to Splunk

Figure 16. Host IP address and Port Number Assignment

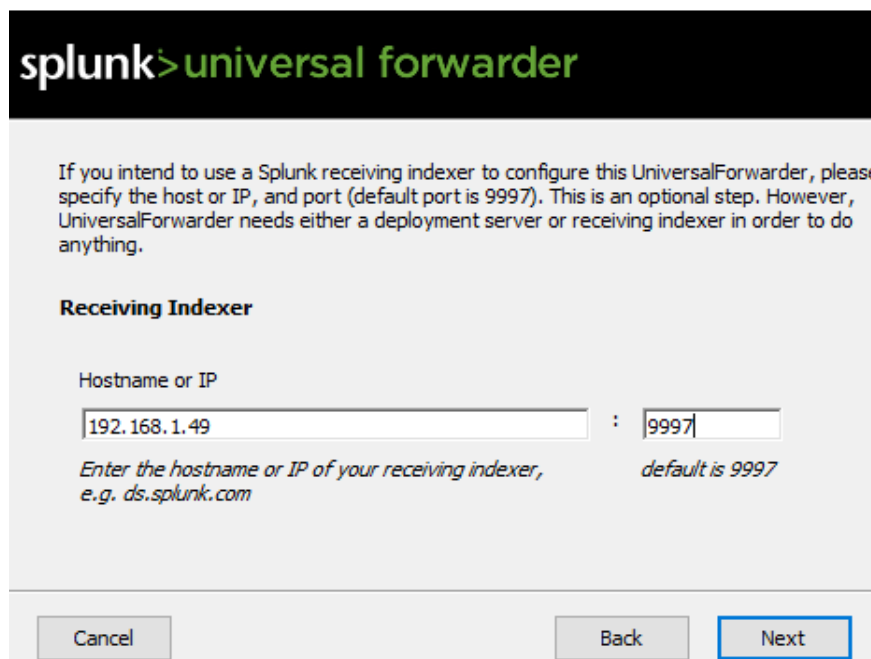


Figure 17. Successfully downloaded Splunk Universal Forwarder

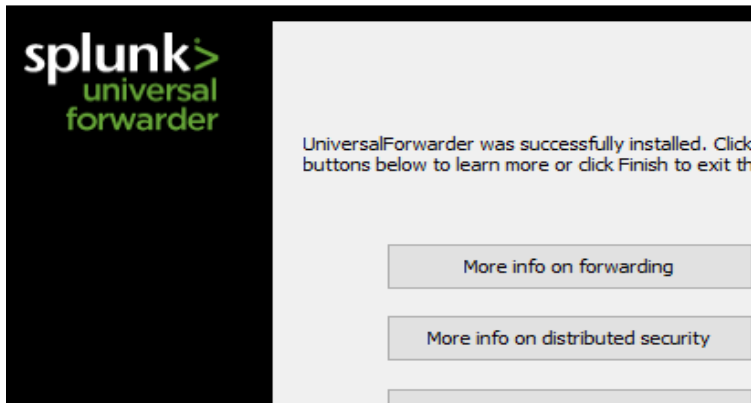


Figure 18. Testing the connection using command line

```
PS C:\Program Files\splunk\bin> .\splunk display listen
WARNING: Server Certificate Hostname Validation is disabled.
details.
Splunk username: SOC-Prakash
Password:
Receiving is enabled on port 9997.
PS C:\Program Files\splunk\bin> █
```

8. Creating input.conf file for events

Figure 19. Inputs file creation with four categories

```
File Edit Format View Help
[WinEventLog://Application]
disabled = 0

[WinEventLog://Security]
disabled = 0

[WinEventLog://System]
disabled = 0

[WinEventLog://Microsoft-Windows-Sysmon/Operational]
disabled = 0
renderXml=true
checkpointInterval = 5
```

Figure 20. Resetting the service to ensure configs work

```
Try the new cross-platform PowerShell https://aka.ms/powershell
PS C:\Windows\system32> net stop splunkforwarder
The SplunkForwarder service was stopped successfully.

PS C:\Windows\system32> net start splunkforwarder
The SplunkForwarder service is starting.....
The SplunkForwarder service was started successfully.

PS C:\Windows\system32>
```

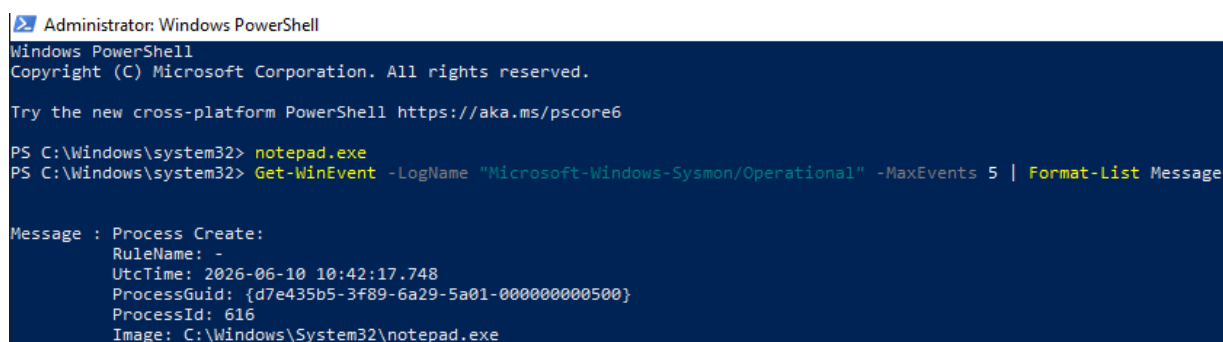
Findings and Observations

After configuring Sysmon and Splunk successfully, I was able to see system activity from the virtual machine inside Splunk.

Activity Performed

I opened Notepad.exe on the Windows virtual machine.

Figure 21. Looking up the event on PowerShell



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> notepad.exe
PS C:\Windows\system32> Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" -MaxEvents 5 | Format-List Message

Message : Process Create:
          RuleName: -
          UtcTime: 2026-06-10 10:42:17.748
          ProcessGuid: {d7e435b5-3f89-6a29-5a01-000000000500}
          ProcessId: 616
          Image: C:\Windows\System32\notepad.exe
```

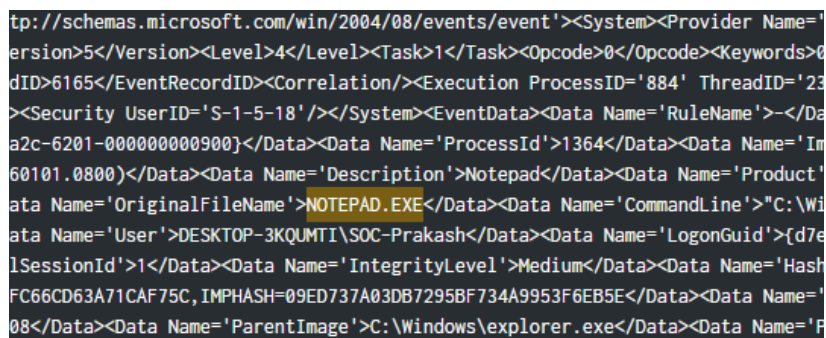
What I Saw in Splunk

Splunk displayed a Sysmon Process Creation event (Event ID 1).

The event showed:

- Process Name: notepad.exe
- User account
- Process ID
- Parent process
- Time of execution

Figure 22. Raw xml for 'notepad.exe'



```
tp://schemas.microsoft.com/win/2004/08/events/event"><System><Provider Name='
ersion>5</Version><Level>4</Level><Task>1</Task><Opcode>0</Opcode><Keywords>0
dID>6165</EventRecordID><Correlation/><Execution ProcessID='884' ThreadID='23
><Security UserID='S-1-5-18'/></System><EventData><Data Name='RuleName'>-</Da
a2c-6201-000000000900}</Data><Data Name='ProcessId'>1364</Data><Data Name='Im
60101.0800)</Data><Data Name='Description'>Notepad</Data><Data Name='Product'
ata Name='OriginalFileName'>NOTEPAD.EXE</Data><Data Name='CommandLine'>"C:\Wi
ata Name='User'>DESKTOP-3KQUMTI\SOC-Prakash</Data><Data Name='LogonGuid'>{d7e
lSessionId'>1</Data><Data Name='IntegrityLevel'>Medium</Data><Data Name='Hash
FC66CD63A71CAF75C,IMPHASH=09ED737A03DB7295BF734A9953F6EB5E</Data><Data Name='
08</Data><Data Name='ParentImage'>C:\Windows\explorer.exe</Data><Data Name='P
```

This confirmed that Sysmon was recording events correctly and that the Universal Forwarder was successfully sending the logs to Splunk.

I was able to see how even a simple action such as opening Notepad generates valuable information that could be used during an investigation.

I also observed other types of events, including:

- Process creation events
- Network connection events
- Registry events
- Process termination events

These events provided a good overview of what was happening on the system.

Figure 23. Raw xml for 'PowerShell.exe'

```
em><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770  
0</Opcode><Keywords>0x8000000000000000</Keywords><TimeC  
ssID='884' ThreadID='2300' /><Channel>Microsoft-Windows-S  
ata Name='RuleName'>-</Data><Data Name='UtcTime'>2026-06  
'>1520</Data><Data Name='Image'>C:\Windows\System32\Wind  
a Name='Description'>Windows PowerShell</Data><Data Name  
OriginalFileName'>PowerShell.EXE</Data><Data Name='Comma  
SOC-Prakash\</Data><Data Name='User'>DESKTOP-3KQUMTI\SOC  
ata><Data Name='TerminalSessionId'>1</Data><Data Name='I  
2776F553C632E58C45E56A0E4639DFD54EE97EAB9C19,IMPHASH=E30  
ata Name='ParentProcessId'>3408</Data><Data Name='Parent
```

Example Splunk Searches

Figure 24. Search results for Sysmon Logs

index=* sourcetype=XmlWinEventLog:Microsoft-Windows-Sysmon/Operational

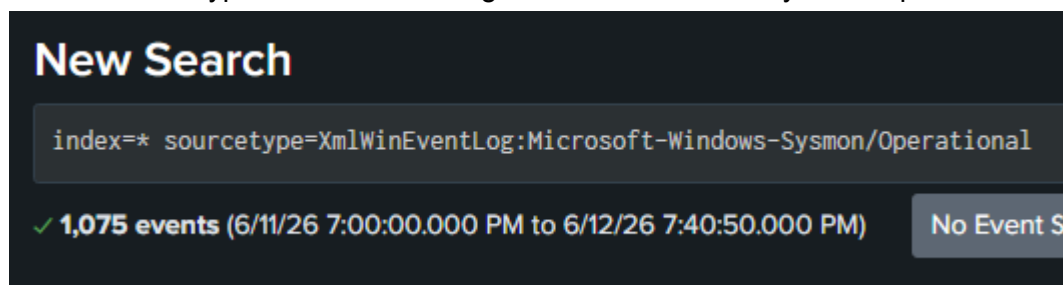


Figure 25. Search results for Notepad

index=* notepad.exe

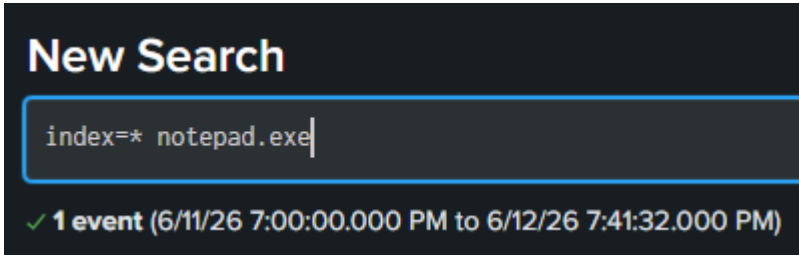


Figure 26. Raw xml for 'Notepad.exe'

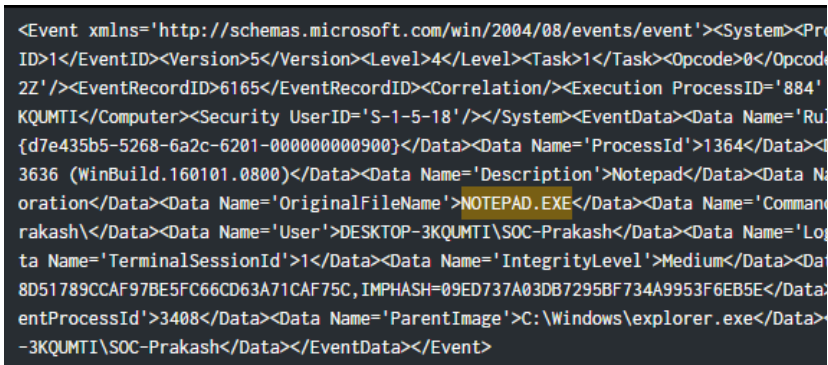


Figure 27. Search results for Process Creation Events

index=* EventCode=1

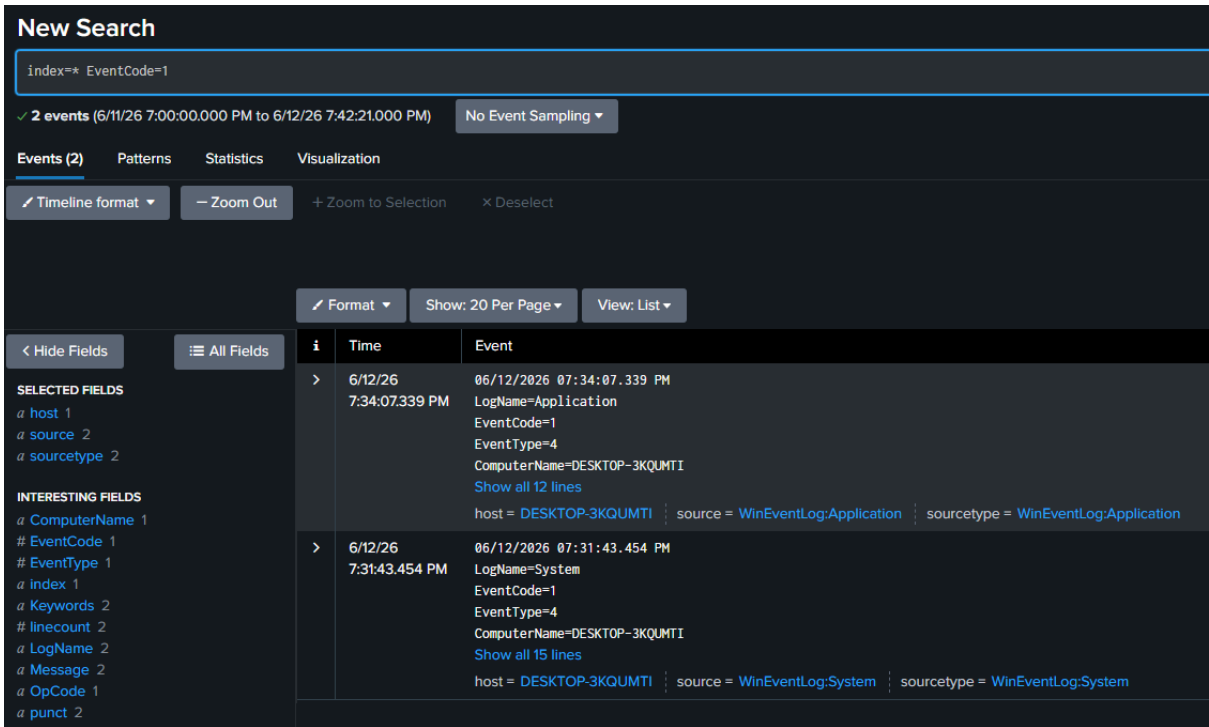
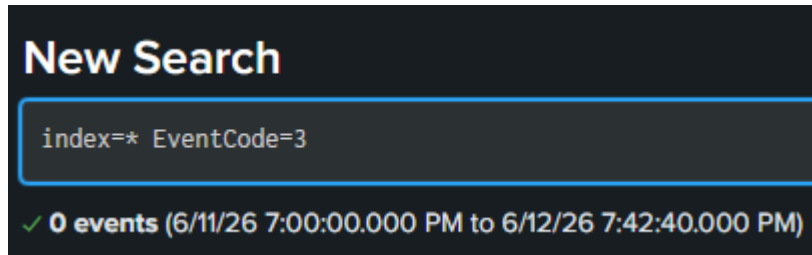


Figure 28. Search results for network connection events

index=* EventCode=3



What I Learned

From this project I learned:

- How Sysmon generates detailed security logs.
- How the SwiftOnSecurity configuration improves Sysmon logging.
- How the Splunk Universal Forwarder sends logs to Splunk.
- How Splunk indexes and stores log data.
- How to perform basic log searches using SPL.
- How SOC analysts investigate endpoint activity.
- The importance of monitoring process creation events.
- How SIEM platforms help security teams detect suspicious behaviour.

This project gave me valuable practical experience and helped me better understand how security monitoring works.

Challenges Faced

During this project, I encountered several challenges while setting up and configuring the environment. Although some issues took a significant amount of time to resolve, they helped me gain a better understanding of how Sysmon, Splunk, and log forwarding work together.

Configuring the Splunk Universal Forwarder

The most challenging part of the project was configuring the Splunk Universal Forwarder on the Windows 10 virtual machine. During installation, I experienced several issues that required multiple reinstalls before the forwarder worked correctly.

One issue I encountered was that the expected log files and directories were not created properly after installation. This prevented logs from being forwarded to Splunk Enterprise and made troubleshooting difficult. After researching the problem and reinstalling the

Universal Forwarder several times, I was eventually able to get the service running correctly and forwarding data.

This challenge helped me better understand how the Universal Forwarder operates and the importance of verifying services, configurations, and installation files during deployment.

Troubleshooting Log Forwarding Issues

Another issue occurred when my host machine's IP address changed. Because the Universal Forwarder had been configured to send logs to the previous IP address, Sysmon events stopped appearing in Splunk.

To resolve the problem, I had to identify the new host IP address and update the forwarder configuration settings. This experience showed me how dependent log forwarding is on correct network configurations and reinforced the importance of checking connectivity when troubleshooting data ingestion problems.

Learning SPL Search Queries

As I was new to Splunk, learning the Search Processing Language (SPL) was initially challenging. At first, I found it difficult to understand how searches were structured and how different fields could be used to locate specific events.

To improve my understanding, I spent time reading Splunk documentation, researching examples online, and reviewing community forum discussions. Over time, I became more comfortable searching for Sysmon events, filtering results, and investigating specific activities such as process creation events.

This helped me develop a better understanding of how analysts use Splunk during security investigations.

Understanding Indexes and Data Parsing

I also experimented with creating multiple indexes to organise the data being collected. Although the logs were successfully ingested, many of the events were still displayed in a raw format, making them more difficult to analyse.

Through further research, I learned that additional Splunk add-ons and field extractions are often required to properly parse and structure Windows and Sysmon logs. Looking back, I believe installing a dedicated Splunk add-on for Sysmon may have improved the presentation and usability of the data.

Although I was not able to fully resolve this issue during the project, it helped me understand the difference between simply collecting logs and properly parsing and normalising them for analysis.

Reflection

While several challenges were encountered throughout the project, each one provided an opportunity to improve my troubleshooting, research, and problem-solving skills. Working through these issues gave me a much better understanding of how Sysmon, Splunk Enterprise, and the Universal Forwarder interact within a security monitoring environment and provided valuable hands-on experience that cannot be gained through theory alone.

References

Microsoft Sysmon Documentation

<https://learn.microsoft.com/sysinternals/downloads/sysmon>

Splunk Documentation

<https://docs.splunk.com>

Splunk Universal Forwarder Documentation

<https://docs.splunk.com/Documentation/Forwarder>

SwiftOnSecurity Sysmon Configuration

<https://github.com/SwiftOnSecurity/sysmon-config>